Neeraj Gurunathan

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Education

BITS Pilani, Dubai Campus Sept. 2024 – June 2028
Bachelor of Engineering in Mathematics and Computing *Dubai, UAE*

- CGPA: 8.98/10
- Relevant Coursework:Deep Learning, Mathematical Modeling Stochastic Calculus & Appl. to Finance, Probability & Statistics, Linear Algebra, OOP, DSA, Numerical Analysis, Numerical Optimization

Experience

Indian Institute of Technology (IIT) Roorkee June 2026 – July 2026
Research Intern – ML-Based Cost Overrun Prediction *Roorkee, India*

- Engineered geospatial and climate feature pipelines in Python using GIS, IMD, CHIRPS, and CHIRTS datasets, processing ∼10,000 data points.
- Performed feature correlation analysis to identify key environmental and project-level drivers of infrastructure cost overruns.
- Trained, tuned, and benchmarked multiple supervised ML models to predict cost overruns and identify the best-performing approach.
- Integrated Monte Carlo simulations with the optimal model to quantify prediction uncertainty and classify project-level financial risk.

AGCOE 2026 – Present
AI Research Publication Coordinator *BITS Pilani, Dubai Campus*

- Coordinate AI research publications, supporting literature reviews, technical research development, manuscript preparation, and peer-review workflows.
- Collaborate with researchers to evaluate and refine technical work, synthesize emerging AI research, and facilitate the dissemination of high-quality research outputs.

Projects

Crypto Options Volatility Modelling | *Python, Deribit API, NumPy, SciPy, Matplotlib* 2026

- Developed rough Bergomi and rough Heston option pricers from scratch in Python to test for rough volatility in BTC/ETH.
- Built a fractional Brownian motion simulator via BLP hybrid scheme and solved fractional Riccati equations.
- Estimated Hurst parameter via Gatheral-Jaisson-Rosenbaum method; verified robustness using Fukasawa's approach.
- Calibrated SSVI, SABR, Heston, and rough Bergomi models against live Deribit order book data.

Professor-Supervised Mathematical Research | *Stochastic Particle Dynamics* Feb. 2026 – Present

- Developing numerical Monte Carlo solvers in Python for coagulation-fragmentation and Becker-Döring equations to model steady-state cluster distributions and their applications in financial markets.

Achievements & Certifications

- **Finalist, IMC Prosperity (2026):** Ranked #1 Nationally in Manual Trading, deploying quantitative and analytical problem-solving under time-sensitive, competitive conditions; placed 3275th of 18803 teams globally.
- **QRT Asset Allocation Performance Forecasting Challenge (2026):** Achieved a score of 0.5158 against a benchmark of 0.50 (M2PF Data Challenge, Ecole Polytechnique), ranking in the top 10 of the private academic leaderboard.
- **Certification:** Introduction to Machine Learning for Trading – QuantInsti (2026)

Technical Skills

Programming Languages: Python (NumPy, SciPy, Pandas, scikit-learn, Matplotlib, PyTorch), R, C++, C
Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, LaTeX
Core Domains: Quantitative Finance Research, Stochastic Processes, Machine Learning, Time Series Analysis, AI Research